12

The Regression Line

The estimation of a magnitude using an observation subject to a larger or smaller error can be compared not inappropriately to a game of chance in which one can only lose and never win and in which each possible error corresponds to a loss . . . . However, what specific loss we should ascribe to any specific error is by no means clear of itself. In fact, the determination of this loss depends at least in part on our judgment . . . . Among the infinite variety of possible functions the one that is simplest seems to have the advantage and this is unquestionably the square . . . . Laplace treated the problem in a similar fashion, but he chose the size of the error as the measure of loss. However, unless we are mistaken this choice is surely not less arbitrary than ours.

—C. F. GAUSS (GERMANY, 1777–1855)[1]

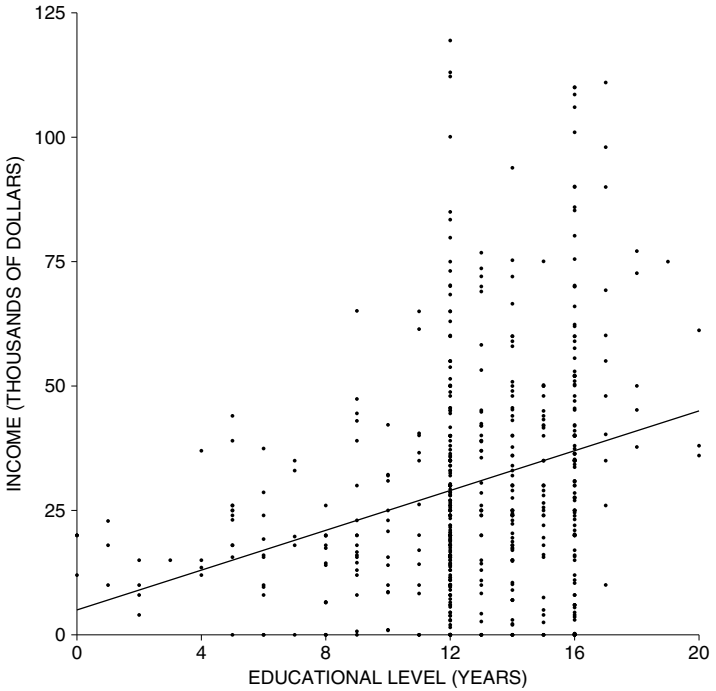
1. SLOPE AND INTERCEPT

Does education pay? Figure 1 shows the relationship between income and education, for a sample of 562 California men age 25–29 in 2005. The summary statistics:[2]

average education $\approx$ 12.5 years, SD $\approx$ 3 years
average income $\approx$ \$30,000, SD $\approx$ \$24,000, $r \approx 0.25$

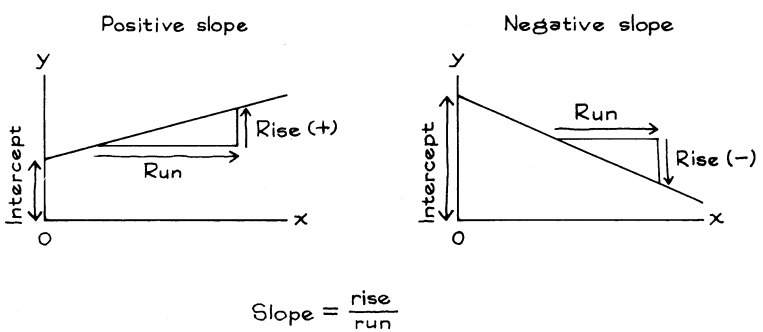
The regression estimates for average income at each educational level fall along the regression line shown in the figure. The line slopes up, showing that on the average, income does go up with education.

Figure 1. The regression line. The scatter diagram shows income and education, for a sample of 562 California men age 25–29 in 2005.



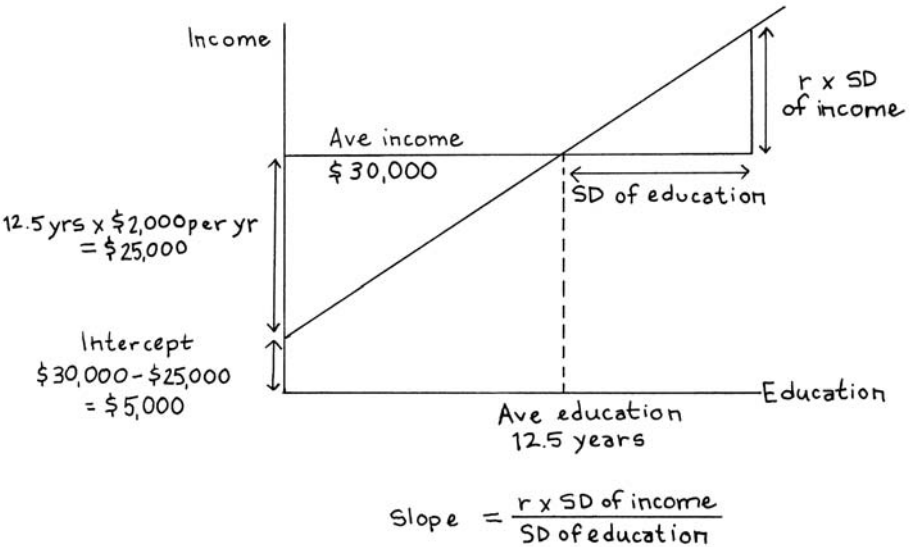
Any line can be described in terms of its slope and intercept (chapter 7). The y-intercept is the height of the line when x is 0. And the slope is the rate at which y increases, per unit increase in x. Slope and intercept are illustrated in figure 2.

Figure 2. Slope and intercept.



How do you get the slope of the regression line? Take the income-education example. Associated with an increase of one SD in education, there is an increase of r SDs in income. On this basis, 3 extra years of education are worth an extra

Figure 3. Finding the slope and intercept of the regression line.



$0.25 \times \$24{,}000 = \$6{,}000$ of income, on the average. So each extra year is worth $\$6{,}000/3 = \$2{,}000$. The slope of the regression line is \$2,000 per year.

The intercept of the regression line is the height when $x = 0$, corresponding to men with 0 years of education. These men are 12.5 years below average in education. Each year costs \$2,000—that is what the slope says. A man with no education should have an income which is below average by

$$12.5 \text{ years} \times \$2{,}000 \text{ per year} = \$25{,}000.$$

His income should be $\$30{,}000 - \$25{,}000 = \$5{,}000$. That is the intercept (figure 3): the predicted value of y when $x = 0$. Zero years of education may sound extreme, but there were four men who reported having no education; their points are in the lower left corner of figure 1. (Three of these men reported incomes of \$20,000, so their points plot one on top of the other.)

> Associated with a unit increase in x there is some average change in y. The slope of the regression line estimates this change. The formula for the slope is
>
> $$\frac{r \times \text{SD of } y}{\text{SD of } x}$$
>
> The intercept of the regression line is just the predicted value for y when x is 0.

The equation of a line can be written in terms of the slope and intercept:

$$y = \text{slope} \times x + \text{intercept}.$$

The equation for the regression line is called (not surprisingly) the *regression equation.* In figure 3, the regression equation is

$$\text{predicted income} = \$2{,}000 \text{ per year} \times \text{education} + \$5{,}000.$$

There is nothing new here. The regression equation is just a way of predicting y from x by the regression method. However, social scientists often report the regression equation; the slope and intercept can be interesting in their own right.

Example 1. Education and income for 570 California women age 25–29 are shown in figure 9 on p. 192. The summary statistics are:[3]

$$\begin{aligned} \text{average education} &\approx 13.0 \text{ years}, & \text{SD} &\approx 3.4 \text{ years} \\ \text{average income} &\approx \$18{,}000, & \text{SD} &\approx \$20{,}000, \quad r \approx 0.37 \end{aligned}$$

(a) Find the regression equation for predicting income from education.

(b) Use the equation to predict the income of a woman whose educational level is: 8 years, 12 years, 16 years.

Solution. Part (a). The first step is to find the slope. In a run of one SD of education, the regression line rises r SDs of income. So

$$\text{slope} = \frac{0.37 \times \$20{,}000}{3.4 \text{ years}} \approx \$2{,}176 \text{ per year}.$$

On the average, each extra year of schooling is worth an extra \$2,176 of income; each year less of schooling costs \$2,176 of income. (Income has such a large SD because the distribution has a long right hand tail.)

The next step is to find the intercept. That is the height of the regression line at $x = 0$—in other words, the predicted income for a woman with no education. She is 13 years below average. Her income should be below average by

$$13 \text{ years} \times \$2{,}176 \text{ per year} = \$28{,}288.$$

Her predicted income is

$$\$18{,}000 - \$28{,}288 = -\$10{,}288.$$

That is the intercept: the prediction for y when $x = 0$. The regression equation is

$$\text{predicted income} = \$2{,}176 \text{ per year} \times \text{education} - \$10{,}288.$$

The regression line becomes unreliable when you are far from the center of the data, so a negative intercept is not too disturbing.

Part (b). Substitute 8 years for education, to get

$$\$2{,}176 \text{ per year} \times 8 \text{ years} - \$10{,}288 = \$7{,}120.$$

Substitute 12 years for education:

$$\$2,176 \text{ per year} \times 12 \text{ years} - \$10,288 = \$15,824.$$

Substitute 16 years:

$$\$2,176 \text{ per year} \times 16 \text{ years} - \$10,288 = \$24,528.$$

This completes the solution. Despite the negative intercept, the predictions are quite reasonable for most of the women.

In example 1, the slope is \$2,176 per year. Associated with each extra year of education, there is an increase of \$2,176 in income, on the average. The phrase "associated with" sounds like it is talking around some difficulty, and here is the issue. Are income differences caused by differences in educational level, or do both reflect the common influence of some third variable? The phrase "associated with" was invented to let statisticians talk about regressions without having to commit themselves on this sort of point.

The slope is often used to predict how y will respond, if someone intervenes and changes x. This is legitimate when the data come from a controlled experiment. With observational studies, the inference is often shaky—because of confounding. Look at example 1. On the average, the women who finished high school (12 years of education) earned about \$9,000 more than women who just finished middle school (8 years).

Now, take a representative group of women with 8 years of education. If the government intervened and sent them on to get high-school degrees, the slope suggests that their incomes would go up by an average of $4 \times \$2,176 \approx \$9,000$. However, example 1 is based on survey data rather than a controlled experiment. One group of women in the survey had 8 years of education. Another, separate, group had 12 years. The two groups were different with respect to many factors besides education—like intelligence, ambition, and family background.

The effects of these factors are confounded with the effect of education, and go into the slope. Sending people off to high school probably would make their incomes go up, but not by the full \$9,000. To measure the impact on incomes, it might be necessary to run a controlled experiment. (Many investigators would use a technique called *multiple regression*; more about this in section 3.[4])

With an observational study, the slope and intercept of the regression line are only descriptive statistics. They say how the average value of one variable is related to values of another variable in the population being observed. The slope cannot be relied on to predict how y would respond if you intervene to change the value of x.

> If you run an observational study, the regression line only describes the data that you see. The line cannot be relied on for predicting the results of interventions.

There is another assumption that we have been making throughout this section: that the average of y depends linearly on x. If the relationship is non-linear, the

regression line may be quite misleading—whether the data come from an experiment or an observational study.[5]

Exercise Set A

1. For the men in figure 1, the regression equation for predicting average income from education is

$$\text{predicted income} = \$2{,}000 \text{ per year} \times \text{education} + \$5{,}000.$$

   Predict the income for one of these men who has

   (a) 8 years of schooling—elementary education
   (b) 12 years of schooling—a high-school diploma
   (c) 16 years of schooling—a college degree.

2. The International Rice Research Institute in the Philippines developed the hybrid rice IR 8, setting off "the green revolution" in tropical agriculture. Among other things, they made a thorough study of the effects of fertilizer on rice yields. These experiments involved a number of experimental plots (of about 20 square yards in size). Each plot was planted with IR 8, and fertilized with some amount of nitrogen chosen by the investigators. (The amounts ranged from 0 to about a pound.) When the rice was harvested, the yield was measured and related to the amount of nitrogen applied. In one such experiment, the correlation between rice yield and nitrogen was 0.95, and the regression equation was[6]

$$\text{predicted rice yield} = (20 \text{ oz rice per oz nitrogen}) \times (\text{nitrogen}) + 240 \text{ oz}.$$

   (a) An unfertilized plot can be expected to produce around _____ of rice.
   (b) Each extra ounce of nitrogen fertilizer can be expected to increase the rice yield by _____ .
   (c) Predict the rice yield when the amount of fertilizer is

   3 ounces of nitrogen 4 ounces of nitrogen

   (d) Was this an observational study or a controlled experiment?
   (e) In fact, fertilizer was applied only in the following amounts: 0 ounces, 4 ounces, 8 ounces, 12 ounces, 16 ounces. Would you trust the prediction for 3 ounces of nitrogen, even though this particular amount was never applied?
   (f) Would you trust the prediction for 100 ounces of nitrogen?

3. Summary statistics for heights of fathers and sons are on p. 170.

   (a) Find the regression equation for predicting the height of a son from the height of his father.
   (b) Find the regression equation for predicting the height of a father from the height of his son.

4. An expert witness offers testimony that[7]

   > Regression is a substitute for controlled experiments. It provides a precise estimate of the effect of one variable on another.

   Comment briefly.

The answers to these exercises are on pp. A65–66.

2. THE METHOD OF LEAST SQUARES

Chapter 10 discussed regression from one point of view, and section 1 went over the same ground using the regression equation. This section is a third pass at the same topic, from yet another perspective. (For statisticians, regression is an important technique.) Sometimes the points on a scatter diagram seem to be following a line. The problem discussed in this section is how to find the line which best fits the points. Usually, this involves a compromise: moving the line closer to some points will increase its distance from others. To resolve the conflict, two steps are necessary. First, define an average distance from the line to all the points. Second, move the line around until this average distance is as small as possible.

To be more specific, suppose the line will be used to predict y from x. Then, the error made at each point is the vertical distance from the point to the line. In statistics, the usual way to define the average distance is by taking the root-mean-square of the errors. This measure of average distance is called the *r.m.s. error of the line*. (It was first proposed by Gauss; see the chapter opening quote.)

The second problem, how to move the line around to minimize the r.m.s. error, was also solved by Gauss.

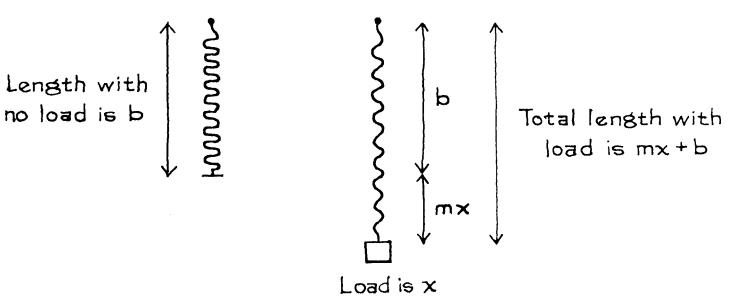
> Among all lines, the one that makes the smallest r.m.s. error in predicting y from x is the regression line.

For this reason, the regression line is often called the *least squares line*: the errors are squared to compute the r.m.s. error, and the regression line makes the r.m.s. error as small as possible. (The r.m.s. error of the regression line was discussed in section 1 of chapter 11.)

Now, an example. Robert Hooke (England, 1653–1703) was able to determine the relationship between the length of a spring and the load placed on it. He just hung weights of different sizes on the end of a spring, and watched what happened. When he increased the load, the spring got longer. When he reduced the load, the spring got shorter. And the relationship was linear.

Let b be the length of the spring with no load. A weight of x kilograms is attached to the end of the spring. As illustrated in figure 4, the spring stretches to

Figure 4. Hooke's law: the stretch is proportional to the load.



a new length. According to Hooke's law, the amount of stretch is proportional to the weight x. The new length of the spring is

$$y = mx + b.$$

In this equation, m and b are constants which depend on the spring. Their values are unknown, and have to be estimated using experimental data.

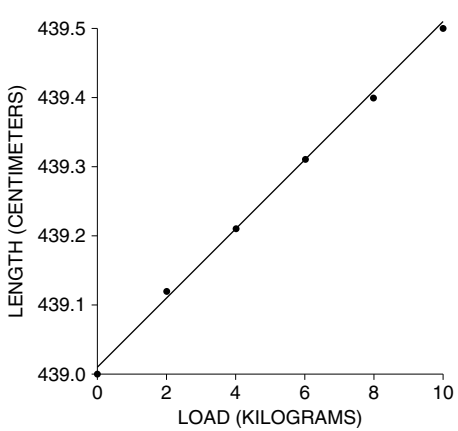
Table 1 shows the results of an experiment on Hooke's law, done in a physics class at the University of California, Berkeley. Different weights were hung on the end of a long piano wire.[8] The first column shows the load. The second column shows the measured length. With 20 pounds of load, this "spring" only stretched about 0.2 inch (10 kg $\approx$ 22 lb, 0.5 cm $\approx$ 0.2 in). Piano wire is not very stretchy.

Table 1. Data on Hooke's law.

Weight (kg)	*Length (cm)*
0	439.00
2	439.12
4	439.21
6	439.31
8	439.40
10	439.50

The correlation coefficient for the data in table 1 is 0.999, very close to 1 indeed. So the points almost form a straight line (figure 5), just as Hooke's law predicts. The minor deviations from linearity are probably due to measurement error; neither the weights nor the lengths have been measured with perfect accuracy. (Nothing ever is.)

Figure 5. Scatter diagram for table 1.



Our goal is to estimate m and b in the equation of Hooke's law for the piano wire:

$$y = mx + b.$$

The graph of this equation is a perfect straight line. If the points in figure 5 happened to fall exactly on some line, the slope of that line would estimate m, and its intercept would estimate b. However, the points do not line up perfectly. Many different lines could be drawn across the scatter diagram, each having a slightly different slope and intercept.

Which line should be used? Hooke's equation predicts length from weight. As discussed above, it is natural to choose m and b so as to minimize the r.m.s. error: this is the *method of least squares*. The line $y = mx + b$ which does the job is the regression line.[9] In other words, m in Hooke's law should be estimated as the slope of the regression line, and b as its intercept. These are called the *least squares estimates*, because they minimize root-mean-square error. If you do the arithmetic,

$$m \approx 0.05 \text{ cm per kg and } b \approx 439.01 \text{ cm}$$

The length of the spring under no load is estimated as 439.01 cm. And each kilogram of load causes the spring to stretch by about 0.05 cm. There is no need to hedge, because the estimates are based on a controlled experiment. The investigator puts the weights on, and the wire stretches. Take the weights off, and the wire comes back to its original length. This can be repeated as often as you want. There is no question here about what is causing what; the language of "association" is not needed. Of course, even Hooke's law has its limits: beyond some point, the spring will break. *Extrapolating beyond the data is risky.*

The method of least squares and the regression method involve the same mathematics; but the contexts may be different. In some fields, investigators talk about "least squares" when they are estimating parameters—unknown constants of nature like m and b in Hooke's law. In other fields, investigators talk about regression when they are studying the relationship between two variables, like income and education, using non-experimental data.

A technical point: The least squares estimate for the length of the spring under no load was 439.01 cm. This is a tiny bit longer than the measured length at no load (439.00 cm). A statistician might trust the least squares estimate over the measurement. Why? Because the least squares estimate takes advantage of all six measurements, not just one: some of the measurement error is likely to cancel out. Of course, the six measurements are tied together by a good theory—Hooke's law. Without the theory, the least squares estimate wouldn't be worth much.

Exercise Set B

1. For the men age 25–34 in the HANES2 sample (p. 58), the regression equation for predicting height from education is[10]

$$\text{predicted height} = (0.25 \text{ inches per year}) \times (\text{education}) + 66.75 \text{ inches}$$

Predict the height of a man with 12 years of education; with 16 years of education. Does going to college increase a man's height? Explain.

2. For the data in table 1 (p. 209), the regression equation for predicting length from weight is

$$\text{predicted length} = (0.05 \text{ cm per kg}) \times (\text{weight}) + 439.01 \text{ cm}$$

Predict the length of the wire when the weight is 3 kg; 5 kg. Does putting more weight on the spring make it longer? Explain.

3. A study is made of Math and Verbal SAT scores for the entering class at a certain college. The summary statistics:

$$\text{average M-SAT} = 560, \quad \text{SD} = 120$$
$$\text{average V-SAT} = 540, \quad \text{SD} = 110, \quad r = 0.66$$

The investigator uses the SD line to predict V-SAT score from M-SAT score.

(a) If a student scores 680 on the M-SAT, the predicted V-SAT score is _____.
(b) If a student scores 560 on the M-SAT, the predicted V-SAT score is _____.
(c) The investigator's r.m.s. error is _____ $\sqrt{1 - 0.66^2} \times 110$. Options:

greater than equal to less than

If more information is needed, say what you need, and why.

4. Repeat exercise 3, if the investigator always predicts a V-SAT of 540.

5. Exercise 3 describes one way to predict V-SAT from M-SAT; exercise 4 describes a second way; and regression is a third way. Which way will have the smallest r.m.s. error?

The answers to these exercises are on p. A66.

3. DOES THE REGRESSION MAKE SENSE?

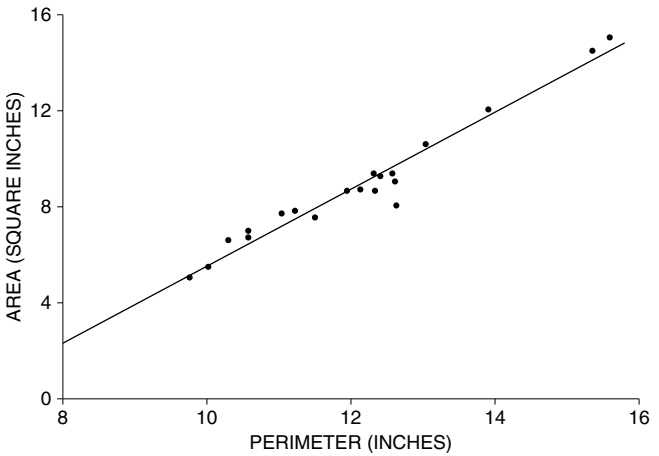
A regression line can be put down on any scatter diagram. However, there are two questions to ask: First, was there a non-linear association between the variables? If so, the regression line may be quite misleading (p. 163). Even if the association looks linear, there is a second question: Did the regression make sense? The second question is harder. Answering it requires some understanding of the mechanism which produced the data. If this mechanism is not understood, fitting a line can be intellectually disastrous.

To make up an example, suppose an investigator does not know the formula for the area of a rectangle. He thinks area ought to depend on perimeter. Taking an empirical approach, he draws 20 typical rectangles, measuring the area and the perimeter for each one. The correlation coefficient turns out to be 0.98—almost as good as Hooke's law. The investigator concludes that he is really on to something. His regression equation is

$$\text{area} = (1.60 \text{ inches}) \times (\text{perimeter}) - 10.51 \text{ square inches}$$

(Area is measured in square inches and perimeter in inches.)

Figure 6. Scatter diagram of area against perimeter for 20 rectangles; the regression line is shown too.



There is a scatter diagram in figure 6, with one dot for each rectangle; the regression line is plotted too. The rectangles themselves are shown in figure 7. The arithmetic is all in order, but the regression is silly. The investigator should have looked at two other variables, length and width. These two variables determine both area and perimeter:

$$\text{area} = \text{length} \times \text{width}, \qquad \text{perimeter} = 2(\text{length} + \text{width})$$

Our straw-man investigator would never find this out by doing regressions.

When looking at a regression study, ask yourself whether it is more like Hooke's law, or more like area and perimeter. Of course, the area-perimeter example is hypothetical. But many investigators do fit lines to data without facing the issues. That can make a lot of trouble.[11]

Technical note. Example 1 in section 1 presented a regression equation for predicting income from education. This is a good way to describe the relationship between income and education. But it may not be legitimate to interpret the slope as the effect on income if you intervene to change education. The problem—the effects of other variables may be confounded with the effects of education.

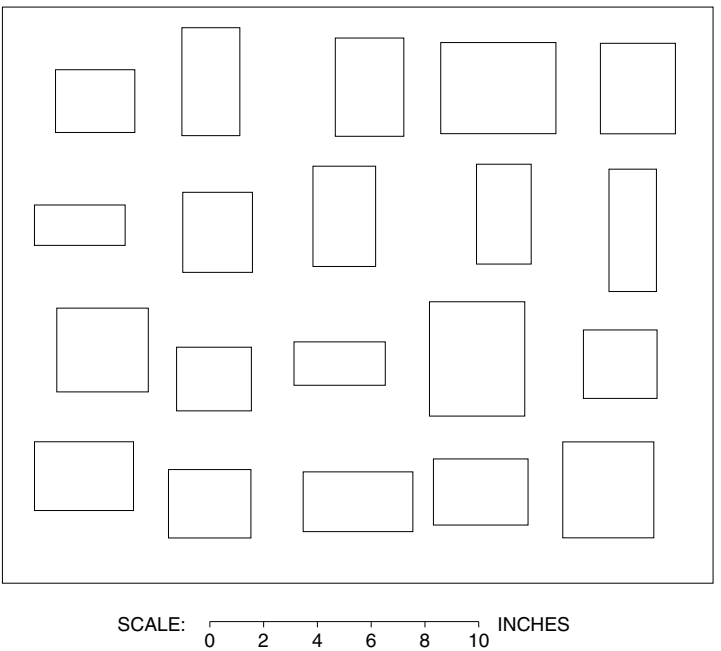
Many investigators would use multiple regression to control for other variables. For instance, they might develop some measure for the socioeconomic status of parents, and fit a multiple regression equation of the form

$$y = a + b \times E + c \times S,$$

where

$$y = \text{predicted income}, \quad E = \text{educational level},$$
$$S = \text{measure of parental status}.$$

Figure 7. The 20 rectangles themselves.



The coefficient b would be interpreted as showing the effect of education, controlling for the effect of parental status.

This might give sensible results. But it can equally well produce nonsense. Take the hypothetical investigator who was working on the area of rectangles. He could decide to control for the shape of the rectangles by multiple regression, using the length of the diagonal to measure shape. (Of course, this isn't a good measure of shape, but nobody knows how to measure status very well either.) The investigator would fit a multiple regression equation of the form

$$\text{predicted area} = a + b \times \text{perimeter} + c \times \text{diagonal}.$$

He might tell himself that b measures the effect of perimeter, controlling for the effect of shape. As a result, he would be even more confused than before. The perimeter and the diagonal do determine the area, but only by a non-linear formula. Multiple regression is a powerful technique, but it is not a substitute for understanding.

4. REVIEW EXERCISES

Review exercises may cover material from previous chapters.

1. Find the regression equation for predicting final score from midterm score, based on the following information:

$$\text{average midterm score} = 70, \quad \text{SD} = 10$$
$$\text{average final score} = 55, \quad \text{SD} = 20, \quad r = 0.60$$

2. For women age 25–34 in the HANES5 sample, the relationship between height and income can be summarized as follows:[12]

$$\text{average height} \approx 64 \text{ inches}, \quad \text{SD} \approx 2.5 \text{ inches}$$
$$\text{average income} \approx \$21{,}000, \quad \text{SD} \approx \$20{,}000, \quad r \approx 0.2$$

What is the regression equation for predicting income from height? What does the equation tell you?

3. For men age 18–24 in the HANES5 sample, the regression equation for predicting height from weight is

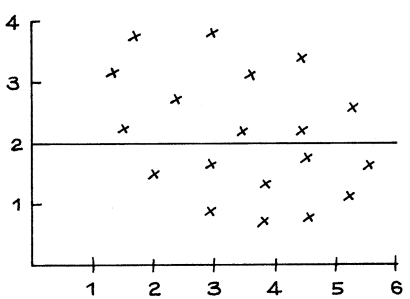
$$\text{predicted height} = (0.0267 \text{ inches per pound}) \times (\text{weight}) + 65.2 \text{ inches}$$

(Height is measured in inches and weight in pounds.) If someone puts on 20 pounds, will he get taller by

$$20 \text{ pounds} \times 0.0267 \text{ inches per pound} \approx 0.5 \text{ inches?}$$

If not, what does the slope mean?

4. (a) Is the r.m.s. error of the line below around 0.1, 0.3, or 1?
(b) Is it the regression line?



5. A study is made of working couples. The regression equation for predicting wife's income from husband's income is

$$\text{wife's income} = 0.1667 \times \text{husband's income} + \$24{,}000.$$

Another investigator solves this equation for husband's income, and gets

$$\text{husband's income} = 6 \times \text{wife's income} - \$144{,}000.$$

True or false, and explain: the second investigator has found the regression equation for predicting husband's income from wife's income. If you want to compute anything,

$$\text{husband's average income} = \$54{,}000, \quad \text{SD} = \$39{,}000$$
$$\text{wife's average income} = \$33{,}000, \quad \text{SD} = \$26{,}000, \quad r = 0.25$$

6. (Continues exercise 5.) The couples in the previous exercise are followed for a year. Suppose everyone's income goes up by 10%. Find the new regression line for predicting wife's income from husband's income.

7. A statistician is doing a study on a group of undergraduates. On average, these students drink 4 beers a month, with an SD of 8. They eat 4 pizzas a month, with an SD of 4. There is some positive association between beer and pizza, and the regression equation is[13]

$$\text{predicted number of beers} = _____ \times \text{number of pizzas} + 2.$$

However, the statistician lost the data and forgot the slope of the equation. (Perhaps he had too much beer and pizza.) Can you help him remember the slope? Explain.

8. An investigator wants to use a straight line to predict IQ from lead levels in the blood, for a representative group of children aged 5–9.[14] There is a weak positive association in the data. True or false, and explain—

   (a) He can use many different lines.
   (b) He has to use the regression line.
   (c) Only the regression line has an r.m.s. error.
   (d) Any line he uses will have an r.m.s. error.
   (e) Among all lines, the regression line has the smallest r.m.s. error.

9. In a large study (hypothetical) of the relationship between parental income and the IQs of their children, the following results were obtained:

$$\text{average income} \approx \$90{,}000, \quad \text{SD} \approx \$45{,}000$$
$$\text{average IQ} \approx 100, \quad \text{SD} \approx 15, \quad r \approx 0.50$$

For each income group ($0–$9999, $10,000–$19,999, $20,000–$29,999, etc.), the average IQ of children with parental income in that group was calculated and then plotted above the midpoint of the group ($5,000, $15,000, $25,000, etc.). It was found that the points on this graph followed a straight line very closely. The slope of this line (in IQ points per dollar) would be about:

6,000 3,000 1,500 500 1/500 1/1,500 1/3,000 1/6,000
can't say from the information given

Explain briefly.

10. One child in the study referred to in exercise 9 had an IQ of 110, but the information about his parents' income was lost. At $150,000 the height of the line plotted in exercise 9 corresponds to an IQ of 110. Is $150,000 a good estimate for the parents' income? Or is the estimate likely to be too high? too low? Explain.

11. (Hypothetical.) A congressional report is discussing the relationship between income of parents and educational attainment of their daughters. Data are

from a sample of families with daughters age 18–24. Average parental income is \$79,300; average educational attainment of the daughters is 12.7 years of schooling completed; the correlation is 0.37.

The regression line for predicting daughter's education from parental income is reported as $y = mx + b$, with x = parental income (dollars), y = predicted education (years), $m = 0.00000925$ years per dollar, and $b = 10.3$ years:

$$\text{predicted education} = 0.00000925 \times \text{income} + 10.3$$

Is anything wrong? Or do you need more information to decide? Explain briefly.

12. Many epidemiologists think that a high level of salt in the diet causes high blood pressure. INTERSALT is often cited to support this view. INTERSALT was a large study done at 52 centers in 32 countries.[15] Each center recruited 200 subjects in 8 age- and sex-groups. Salt intake was measured, as well as blood pressure and several possible confounding variables. After adjusting for age, sex, and the other confounding variables, the authors found a significant association between high salt intake and high blood pressure. However, a more detailed analysis showed that in 25 of the centers, there was a positive association between blood pressure and salt; in the other 27, the association was negative. Do the data support the theory that high levels of salt cause high blood pressure? Answer yes or no, and explain briefly.

5. SUMMARY AND OVERVIEW

1. The regression line can be specified by two descriptive statistics: the *slope* and the *intercept*.

2. The slope of the regression line for y on x is the average change in y, per unit change in x. This equals

$$r \times \text{SD of } y/\text{SD of } x.$$

3. The intercept of the regression line equals the regression estimate for y, when x is 0.

4. The equation of the regression line for y on x is

$$y = \text{slope} \times x + \text{intercept}.$$

5. The equation can be used to make all the regression predictions, by substitution.

6. Among all lines, the regression line for y on x makes the smallest r.m.s. error in predicting y from x. For that reason, the regression line is often called the *least squares line*.

7. Sometimes, two quantities are thought to be connected by a linear relationship (for example, length and weight in Hooke's law). The statistical problem is to estimate the slope and intercept of the line. The *least squares estimates* are the slope and intercept of the regression line.

8. In this part of the book, scatter diagrams are used to graph the association between two variables. If the scatter diagram is football-shaped, it can be summarized by the average and SD of the two variables, with r to measure the strength of the association.

9. How does the average of one variable depend on the values of the other variable? The regression line can be used to answer that question.

10. With a controlled experiment, the slope can tell you the average change in y that would be caused by a change in x. With an observational study, however, the slope cannot be relied on to predict the results of interventions. It takes a lot of hard work to draw causal inferences from observational data, with or without regression.

11. If the average of y depends on x in a non-linear way, the regression line can be quite misleading.